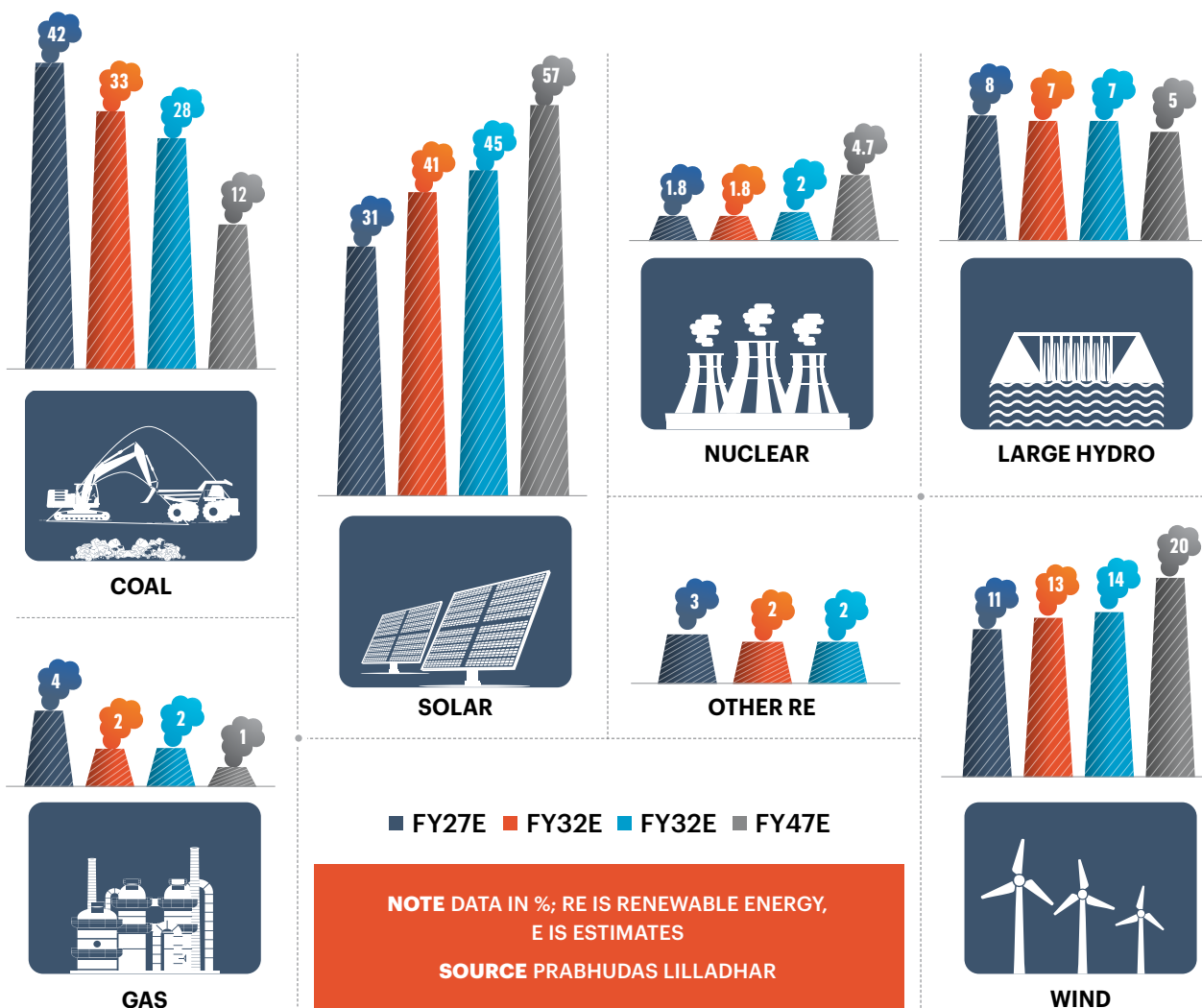


INSTALLED CAPACITY

- India is aiming for 10× expansion of nuclear capacity over the next two decades
- Achieving the 100GW target is expected to cost ₹19 lakh crore



occur in stars and the sun.

Looking at tapping all possible sources of clean energy to meet India's growing power demand, the Central Electricity Authority (CEA), in January this year, constituted a committee for preparing a road map for deployment of nuclear fusion-based power generation.

There has been a massive jump in start-ups working in fusion technology over the past few

years. Globally, over 40 start-ups working on fusion technology have come up since 2024, according to the IAE.

"The reason being, in 2022, the National Ignition Facility at Lawrence Livermore National Laboratory in the US successfully demonstrated Fusion Ignition (generating more energy output than input) for the first time in history. Since then, the race to fusion has significantly heated

up with over 40+ private companies," says Nithish Kumar, Investment Analyst at Speciale Invest. The deep tech VC fund invested in Anubal Fusion in 2024.

"The bottleneck is Uranium 233, which must be bred from thorium and currently depends entirely on the slow FBR build-out. Fusion-fission hybrid technology offers a parallel route: fusion neutrons driving